LESSON 5 — Reading the Joysticks

Values, variables, and the minus sign

MISSION Can you turn a raw FTC Y value into the correct stored power without guessing?

Learning targets

CHECK	BY THE END OF THIS LESSON, I CAN…
[ ]	Read an assignment from right to left.
[ ]	Explain the double data type.
[ ]	Trace both Y-axis inputs.
[ ]	Negate sample values and predict motion.

Words to know

VOCABULARY double | assignment | expression | gamepad1 | Y axis | raw value | unary minus | tank drive

Code focus

```
double leftPower = -gamepad1.left_stick_y;
double rightPower = -gamepad1.right_stick_y;
```

Before the video: commit to a prediction

MY PREDICTION A forward stick usually reports -0.7. What value is stored after the unary minus is applied?

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Confidence before the lesson: 1 2 3 4 5 (circle one)

The POWERING ON safety contract

NON-NEGOTIABLE The sequence is completed before every live-motor test. If a step is skipped, remain disabled and restart at Step 1.

STEP	ACTION	TEAM CHECK
1	SECURE THE ROBOT Use stable blocks. Keep driven wheels and moving mechanisms clear of the table and floor.	[ ]
2	CONFIRM DISABLED Select the correct robot and OpMode, but do not press PLAY.	[ ]
3	LOOK AROUND Remove hands, tools, wires, scraps, and loose objects from moving areas.	[ ]
4	MAKE SURE ALL ARE CLEAR Everyone steps outside the boundary; secure hair, clothing, lanyards, strings, and jewelry.	[ ]
5	CALL AND WAIT Call POWERING ON! loudly. Pause long enough for everyone to stop, look up, and step away.	[ ]
6	VISUALLY CONFIRM, THEN ENABLE Scan the full area again. Press PLAY only when everyone and everything is clear.	[ ]

Callout script

BEFORE PLAY Robot is secured and DISABLED. Look around. Everyone clear? POWERING ON! ... Wait ... Area is visibly clear. Enabling now.

BEFORE CONTACT STOP pressed. Driver Station shows DISABLED. Hands may approach only after the operator confirms disabled.

Student agreement

I understand that anyone may call STOP and the operator will disable immediately. I will not approach or touch the robot until DISABLED is visually confirmed.

Student signature: ______________________________________ **Date:** _______________

Reference program: BasicTwoMotorTeleOp.java

Use this numbered reference throughout Lessons 1-6. The comments divide the program into startup and repeating control sections.

```
package org.firstinspires.ftc.teamcode;

import com.qualcomm.robotcore.eventloop.opmode.LinearOpMode;
import com.qualcomm.robotcore.eventloop.opmode.TeleOp;
import com.qualcomm.robotcore.hardware.DcMotor;

@TeleOp(name="Basic Two Motor TeleOp", group="Iterative Opmode")
public class BasicTwoMotorTeleOp extends LinearOpMode {
  private DcMotor leftMotor;
  private DcMotor rightMotor;

  @Override
  public void runOpMode() {
    leftMotor = hardwareMap.get(DcMotor.class, "left_motor");
    rightMotor = hardwareMap.get(DcMotor.class, "right_motor");

    // Set one motor REVERSE if the physical drivetrain requires it.
    // rightMotor.setDirection(DcMotor.Direction.REVERSE);

    telemetry.addData("Status", "Initialized");
    telemetry.update();
    waitForStart();

    if (opModeIsActive()) {
      while (opModeIsActive()) {
        double leftPower = -gamepad1.left_stick_y;
        double rightPower = -gamepad1.right_stick_y;
        leftMotor.setPower(leftPower);
        rightMotor.setPower(rightPower);
        telemetry.addData("Left Power", leftPower);
        telemetry.addData("Right Power", rightPower);
        telemetry.update();
      }
    }
  }
}
```

CODE DETECTIVE Circle waitForStart(). Box the while condition. Underline both setPower() calls. Draw a star beside each telemetry line.

LESSON 5 — Guided Video Notes

Reading the Joysticks

DIRECTIONS At each timestamp, pause, think silently, discuss, and then record evidence from the code or whiteboard visual.

PAUSE	QUESTION	MY CODE / VISUAL EVIDENCE
0:27	What does double tell Java?	
0:36	Which side of an assignment is calculated first?	
0:46	What does gamepad1 identify, and what does Y measure?	
0:56	How does tank drive use the two sticks?	
1:06	What raw value normally means fully forward?	
1:15	What does the minus sign do to that value?	

MOST IMPORTANT IDEA The idea I would teach to a teammate is:

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LESSON 5	Hands-On Team Activity *Reading the Joysticks*

Hands-On Team Activity

Reading the Joysticks

Materials and setup

READY	ITEM
[ ]	Role cards: RAW VALUE, MINUS SIGN, POWER VARIABLE, and MOTION PREDICTOR.
[ ]	Number cards: -1.0, -0.7, -0.2, 0.0, +0.35, +0.8, and +1.0.
[ ]	Printed joystick-conversion sheet from this guide.
[ ]	BasicTwoMotorTeleOp.java displayed at the leftPower/rightPower assignments.
[ ]	Optional secured robot with power telemetry visible.

Team roles

ROLE	JOB
Left raw value	Displays gamepad1.left_stick_y.
Right raw value	Displays gamepad1.right_stick_y.
Left minus sign	Changes the sign of the left raw value.
Right minus sign	Changes the sign of the right raw value.
Left power	Holds the result stored in leftPower.
Right power	Holds the result stored in rightPower.
Motion predictor	Predicts forward, backward, stop, curve, or spin.
Recorder	Writes raw values, stored values, and predictions.

Activity sequence

STEP	ACTION
1	Display double leftPower = -gamepad1.left_stick_y;. Start at the right side and narrate the order of work.
2	Hand RAW VALUE -0.7 to a student. Ask MINUS SIGN to transform it without changing its magnitude.
3	Give the left and right raw students different cards. Both paths operate at the same time conceptually, but calculate each path clearly.
4	Use raw 0.0 on both sticks. Ask whether the OpMode ends.
5	Secure the robot and complete the POWERING ON sequence. Move one stick gently and compare raw direction, stored sign, and telemetry.

LESSON 5	Lab Record and Exit Ticket
	Reading the Joysticks

Student/team: ______________________________ Date: _______________ Robot: ___________________

BEFORE PLAY	AFTER TEST
[ ] Robot secure [ ] DISABLED [ ] Area clear [ ] POWERING ON called [ ] Wait completed [ ] Visual clear	[ ] STOP pressed [ ] DISABLED visually confirmed Safety lead initials: __________

Joystick sign-conversion practice

RAW Y VALUE	APPLY -Y	STORED POWER	PREDICTED REQUEST
-1.0	-(-1.0)		
-0.7	-(-0.7)		
0.0	-(0.0)		
+0.35	-(+0.35)		
+1.0	-(+1.0)		
Student value: ____			

Debrief

#	QUESTION	MY RESPONSE
1	Why is Y negated?	
2	What does double communicate?	
3	Are leftPower and rightPower permanent fields here?	
4	Does +0.5 telemetry guarantee physical forward motion?	

Exit ticket

ANSWER BEFORE LEAVING If right_stick_y is +0.35, what is stored in rightPower, and what motion does that side request under the lesson convention?

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